

Deepraj Pandey

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EDUCATION

ASHOKA UNIVERSITY

POST GRADUATE DIPLOMA IN
ADVANCED STUDIES AND RESEARCH IN
COMPUTER SCIENCE
May 2021 | Sonapat, India

ASHOKA UNIVERSITY

BSC. HONS. IN COMPUTER SCIENCE
May 2020 | Sonapat, India

D.A.V. PUBLIC SCHOOL

Grad. May 2017 | Bhubaneswar, India

SKILLS

PROGRAMMING

Current:

Python • C • C++ • Javascript • Typescript
• Dart • Flutter • Vue • Hyperledger
Fabric • $\text{\LaTeX}$

Familiar:

R • Shell • DevOps • Android

Other:

Git • Processing • Google Apps Script
Arduino • Ruby on Rails • TI-BASIC

Systems/Simulators:

Linux • MacOS • AWS • GCP • Gem5
NVMain • Champsim • JOS Kernel

COURSEWORK

Programming Language Design (TA)
Program Analysis*
Operating Systems
Intro Machine Learning
Computer Security
Algorithm Design
Advanced Programming

* Adv elective in first year.

VOLUNTEERING

MLH Local Hack Day | December, 2017
Organised a 12 hour internal hackathon on campus in first semester.

Ashoka Sports Fest | October, 2017
Handled logistics for Agneepath - annual inter-University sports event.

Rotary IWC | Bhubaneswar
400+ hours of volunteering 2014 - 2017

MISCELLANEOUS

Calligraphy Hobbyist
Air Rifle Shooting
Mechanical Keyboard Enthusiast
Hacktoberfest - '17, '18, '19

EXPERIENCE

DEPARTMENT OF COMPUTER SCIENCE | TEACHING ASSISTANT

August 2020 - Present | Ashoka University, India

- TA to **Prof. Manu Awasthi** in the introductory PL course for third year students at Ashoka - **Programming Language Design and Implementation**.
- Helping Professor with course logistics and with designing & grading assignments.
- Holding weekly office hours to cover supplementary material and to help students.

BLOCKCHAIN RESEARCH GROUP | RESEARCH ASSISTANT

May 2020 - Present

- Working with **Prof. Mahavir Jhawar** and built **CovidBloc** - a transparent, privacy-preserving contact tracing system on blockchain.
- Submitted to **SPACE 2020** under short paper category.
- Implemented a framework based on the **Crypto Specification of the Google-Apple Exposure Notifications** from scratch in Dart and TypeScript.
- Built the EN server on Hyperledger Fabric, the web dashboard for health officials (improved framework allows for **offline verification** of patients), & Android app on Flutter (partial contribution).

DEPARTMENT OF COMPUTER SCIENCE | RESEARCH ASSISTANT

May 2019 - Aug 2019 | Ashoka University, India

- Worked with **Prof. Manu Awasthi** on non-volatile memory architecture in hand-held devices.
- Designed an interface to **gem5** to add **arbitrarily deep** cache hierarchies with non-volatile memory (via NVMain) for simulations.
- Code reviewed and added to an in-house memory simulation tool.

OFFICE OF ACADEMIC AFFAIRS | INTERN

February 2019 - April 2019 | Ashoka University, India

- Worked with **Prof. Hemanshu Kumar** and the Office to **design** an algorithm and **build** a system for faculty advisors and the OAA to check course credit requirements of final year students for graduation.
- Code submitted, reviewed, and is currently used by University Faculty.

MYTRAH MOBILITY | SUMMER INTERN

May 2018 - August 2018 | Gurgaon, India

- Worked on the Electric Mobility company's fleet electrification analysis **tool**.
- Parsed and prepared data on **4000** public buses in multiple Indian cities. Code reviewed and pushed to production.
- Led a team of all interns to design a long-term plan for non-profit wing, **EMA**.
- Planned the company's flagship hackathon before end of term.

LEADERSHIP

WOMEN IN COMPUTING SOCIETY | HEAD OF EVENTS

August 2019 - July 2020 | Ashoka University, India

- Initially shared the position, later headed a team of 8. Organised workshops, guest talks, and socials, usually around inclusivity in Tech.
- Took workshops on Git, Docker, Intro to Programming, web technologies since 2017 on behalf of the society.

MLH SUMMER LEAGUE HACKATHONS | MENTOR & JUDGE

May 2020 - August 2020 | Remote

- Judged **20-30** submissions at every hackathon in the MLH Summer League - one online event **every weekend** all summer.
- **Mentored** multiple projects at the hackathons that had 500-700 participants from all over the world.

PROJECTS

Superhuman Browser Extension | Freelance

Reverse engineered Superhuman Email Client & added a feature to send automatic text messages on email opens. Private freelance project.

CNN - Chars74K | Colab

81.4% val accuracy

Learning about CNN by experimenting with the architecture and data.

Garbage Collector | GitHub

Implemented the mark and sweep phases of a stop the world garbage collector for Programming Language Design and Implementation class.

Gina | GitHub

A bot built on first principles for syntax analysis and to learn a new language based on user translations.

DES Implementation | GitHub

A DES implementation in C++ from scratch.

Network Design Cisco Packet Tracer

Designed multiple networks on Cisco Packet Tracer - internetworks with DHCP servers, web hosting, DNS servers, VLAN, NAT enabled routers (private networks) etc.

OS Labs on xv6

Coursework similar to MIT 6.828. Edited kernel code to add physical page allocators, interrupt handlers, system calls, page faults, and context switches.

Trace Driven Cache Simulator | GitHub

Simple, proof of work trace based cache simulator with 32KB L1 caches (4 words).

Distributed Server | GitHub

Distributed, multithreaded TCP server with a periodic concurrency phase that ensures all servers have the same copy of the data sent from clients.

Biology Society Blog

Built the biology society's blog using Jekyll with templating for easy maintenance and post additions. Hosted on GitHub pages.

COMMUNITY

Computer Architecture Reading Group - Summer '19, Spring '20

Google Developer Group, Bhubaneswar: 2015 - Present

Google Developer Day: 2016 (Bhubaneswar), 2019 (Gurgaon)

LINKS

GitHub:// [DeeprajPandey](#)

LinkedIn:// [DeeprajPandey](#)

Academia:// [DeeprajPandey](#)

Twitter:// [@DeeprajPandey](#)

BANJAARA - ANNUAL FEST | HEAD OF TECH TEAM

October 2018 – March 2019 | Ashoka University, India

- Led a team of 6 developers, designers for the annual fest.
- Drove the development of the fest website, a full stack registration portal (Laravel, Heroku, Google Cloud Storage), and hardware projects for fest day (Arduino powered photo booth with a pull switch capture setup).

HACKDAV | FOUNDER

October 2015 – Present | Bhubaneswar, India

- Founded the state's first hackathon for high school students.
- Applicants from across India and 5 other countries.
- Led the team that raised **INR 1,800,000** in sponsorships for 2017.
- Major sponsors include DigitalOcean, Soylent, JetBrains, MLH, Balsamiq, Hackerearth along with Google and Microsoft Developer Groups.

CONFERENCES

INTERNATIONAL SYMPOSIUM FOR COMPUTER ARCHITECTURE

| UNDERGRADUATE WORKSHOP ATTENDEE

June 2019 | Phoenix, USA

- One of the 42 undergrads selected from across the world for the **first undergraduate architecture workshop** at ACM FCRC - ISCA 2019.
- Received funding from UC Davis and a travel grant from Ashoka.
- Mentee under **Dr. Parthasarathy Ranganathan** as a part of the Senior Architect Mentor Matching program at the workshop.

MLH HACKCON VI | GITHUB SCHOLAR

June 2018 | Pennsylvania, USA

- Awarded the GitHub Scholarship for attending the student hackathon organiser conference organised by MLH and GitHub.
- Only student organiser representing a high school hackathon (HackDAV).
- Ideated a hacker exchange program with UMass & UTexas with GitHub.

ISTEM HACKATHON | WINNER

Jan 2019 | IIIT Bangalore, India

- Worked in a team of VI and non-VI participants to build web app that extracts text data from coding tutorial videos for viewers with visual impairments.
- Our team won the hackathon which had 75+ participants including Graduate students and professionals from across the country.

NOTRE DAME LEADERSHIP SEMINARS | PARTICIPANT

July 2016 | Notre Dame, USA

- Only student from India (among the 98 from across the world) selected for a **fully-funded** summer program at the University of Notre Dame.
- Seminar on the ethical concerns of space exploration under **Prof. Don Howard** and **Dr. Jessica Baron**.

PROJECTS

SPLIDWISE | GITHUB LINK

May 2020

- Distributed application to help users split their bills by recording payments.
- Hyperledger Fabric 1.4, NodeJS REST API with OAuth, VueJS, AWS, Certbot SSL.
- Links to **Report**, **Presentation**.

TOPIC CLUSTERING OF ENRON EMAILS | GITHUB LINK

November 2019

- Worked in a team of 3 on the topic modelling problem for Intro to ML.
- Tried different approaches and decided on using Latent Dirichlet Allocation which gave the best results.
- Links to **Source Code**, **Report**, **clustering visualisation**.